



GE VERNOVA

ADMS

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Snapshot Viewer User Guide

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1. Introduction

The Snapshot Viewer is an ADMS Outage Management System (OMS) debugging and disaster recovery tool used when all other options of fixing a problem have been exhausted.

The snapshot viewer application allows you to view the entries of snapshot and journal files in a grid view, modify them, and save the changes. For NetServer, it is possible to apply journal messages up until a specified point in time and then create a new snapshot file with the saved results.

The snapshot viewer also allows an administrator to view and edit dynamics files. Dynamics files are standalone, such that a dynamics file can be modified in isolation without requiring modifications in other files before the change is valid. The snapshot viewer allows an administrator to view any dynamics file, modify any of the contents, and save the file.

For system restoration, the administrator must either copy the files to a separate location or coordinate to stop the run-time application manually before saving the file.

Removing "orphan" dynamics entries (those that no longer link to a valid device) does not require the snapshot viewer. Consider the following example: a cut is placed on a line, but a GIS (model) import occurs where the original line no longer exists in the system. In this case, NetServer is unable to apply the cut. The cut can be deleted and potentially re-applied to the system using a common network view display rather than the Snapshot Viewer.

Note

The Snapshot Viewer is designed for system administrators as a debugging tool. For operation staff to view and make corrections to cleared incidents, use the History Correction display, which is part of OMS display plugin.

2. General Usage

The state of the OMS database is determined by the latest snapshot created by the NetServer and a set of journals. The ADMS Snapshot Viewer highlights errors in the database and allows an analyst or an administrator to view and edit the NetServer snapshots, dynamics, and journal files.

The ADMS Snapshot Viewer provides the analyst with the following functionality:

- Open and view the latest snapshots, dynamics, and journals in sequence starting from the snapshot time.
- View color-coded suspected errors in the journal messages.
- Edit messages to fix errors and solve original problems.
- Examine databases to verify that problems are solved.
- Create a new snapshot that incorporates the changes.
- Export dynamics from the TSV files and convert them into the internal SAV format used in ADMS.

2.1. Starting the Snapshot Viewer Tool

To access the Snapshot Viewer from the Windows Start menu, click Start > Programs > Eterra > Distribution > Tools > Configuration Tools.

This shortcut passes a configuration file to the application by specifying its path in the Target field in the format "[Snapshot Viewer Executable Path] [Snapshot Viewer Configuration XML file name]". You can view and modify this parameter in the Target field of the shortcut's properties dialog box (see the following image).

The default Target field value is `D:\Eterra\Distribution\Tools\bin\SnapshotViewer.exe /CONFIG "%IDMS_ROOT%\ToolsWorkspace\SnapshotViewerConfig.xml"`.

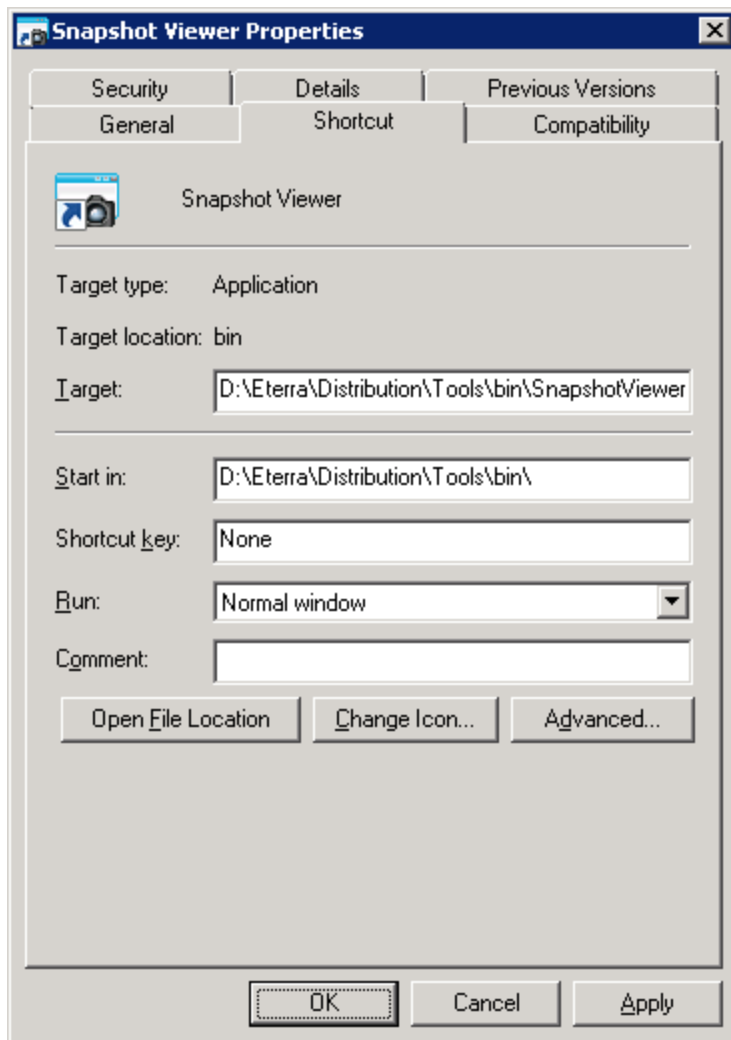


Figure 1. Snapshot Viewer Properties

2.2. Clearing Displays

To clear all the Snapshot Viewer display and reset the application to the initial state:

- Select File > Clear All

2.3. Snapshot Viewer Controls

In the Snapshot Viewer, the File menu includes the following commands:

- **Open:** Opens a snapshot, a journal, or a dynamics file.
- **Create Snapshot:** Creates a new snapshot file (*.snapshot). This is used to save changes after the analyst edits a snapshot file and journals.
- **Create Dynamics:** Creates a new dynamics file (*.sav). This is used to capture the state of the

ADMS dynamics after the analyst reviews and edits the dynamics file. This file can then be uploaded to the ADMS server.

- **Export Dynamics:** Exports the dynamics information (headers and data sections) in the tab-delimited format (*.tsv).
 - **Header Template:** Exports the header information from a SAV file.
 - **Data:** Exports dynamics data from a SAV file.
- **Import Dynamics:** Imports dynamics from a TSV file. To import a dynamics file into the ADMS system (for example, during a system migration) the analyst must import the header file (Templatedynamics.sav). This is required because the TSV file does not contain details about the field data type, field size, etc.
 - **Header Template:** Imports the header information from the Templatedynamics.sav file. The default location of the file is %IDMS_ROOT%\dynamics\server\OMSServer.
 - **Data:** Imports dynamics data from a TSV file.
- **Clear All:** Clears all the data from all the tabs.
- **Export:** Exports the currently opened snapshot or journal file to the Export folder that is created in the directory that contains the currently opened snapshot or journal file.
- **Exit:** Closes the application.

2.4. Displays Overview

The Snapshot Viewer supports multiple tabbed views. The displayed tabs are different for dynamics, journals, and snapshots.

This chapter provides separate overviews of the displays, views, and controls for snapshots, dynamics, and journal files.

2.4.1. Viewing Snapshots

To open a snapshot file, click File > Open > Snapshot. The default location of snapshot files is %IDMS_ROOT%\dynamics\server\OMSServer\Snapshot\.

2.4.1.1. Summary Tab

The Summary tab shows summary information related to the opened snapshot file.

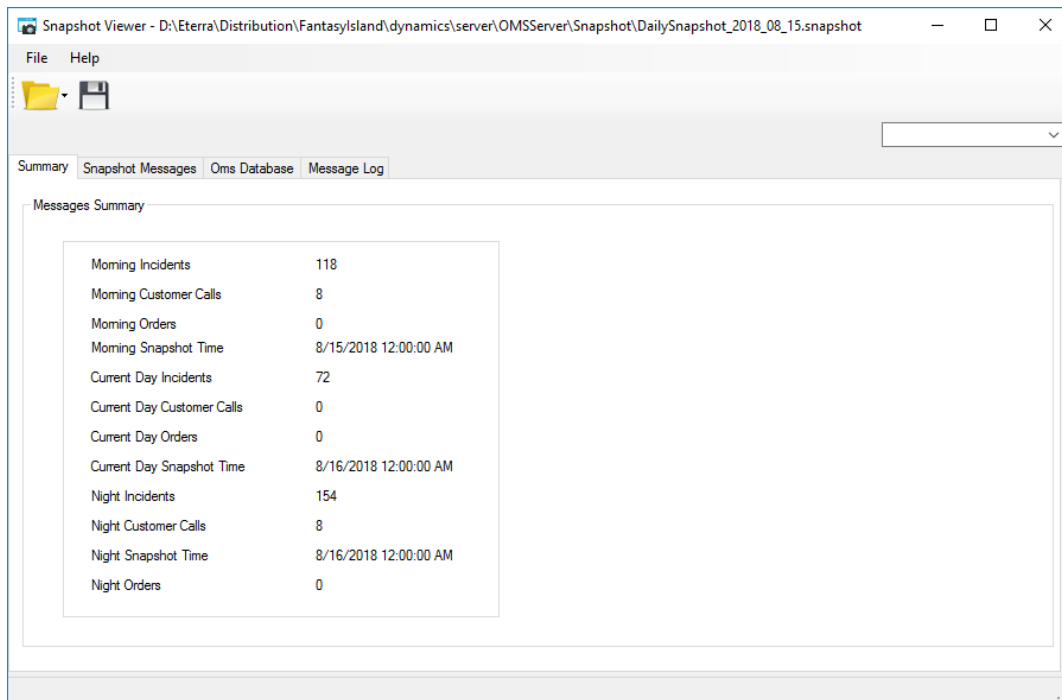


Figure 2. Viewing Daily Snapshots - Summary Tab

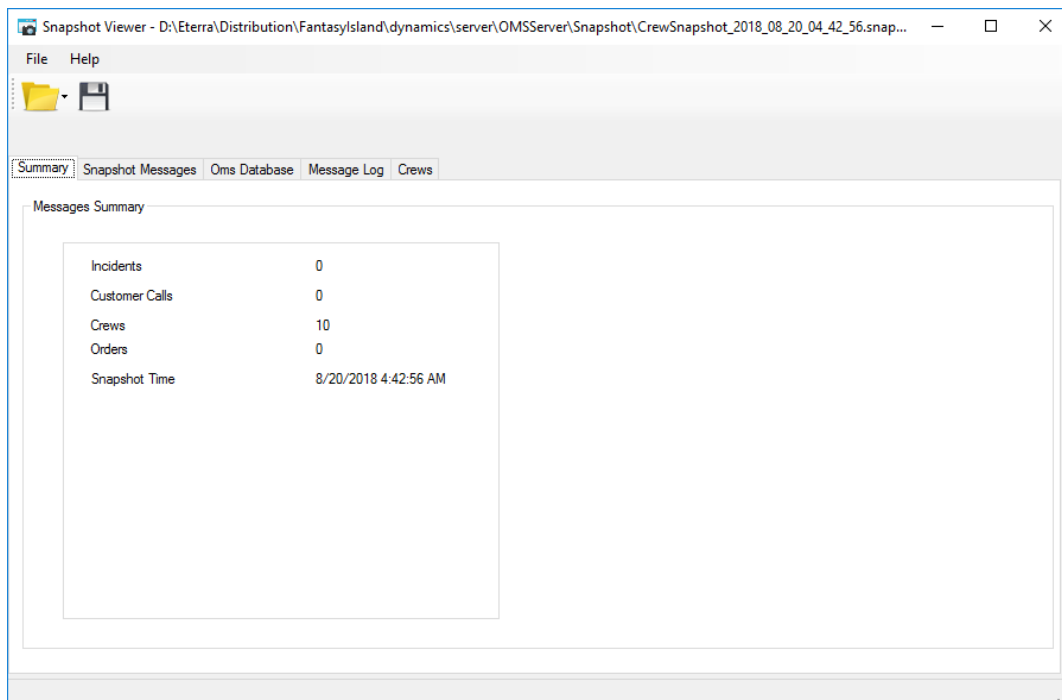


Figure 3. Viewing Crew Snapshots - Summary Tab

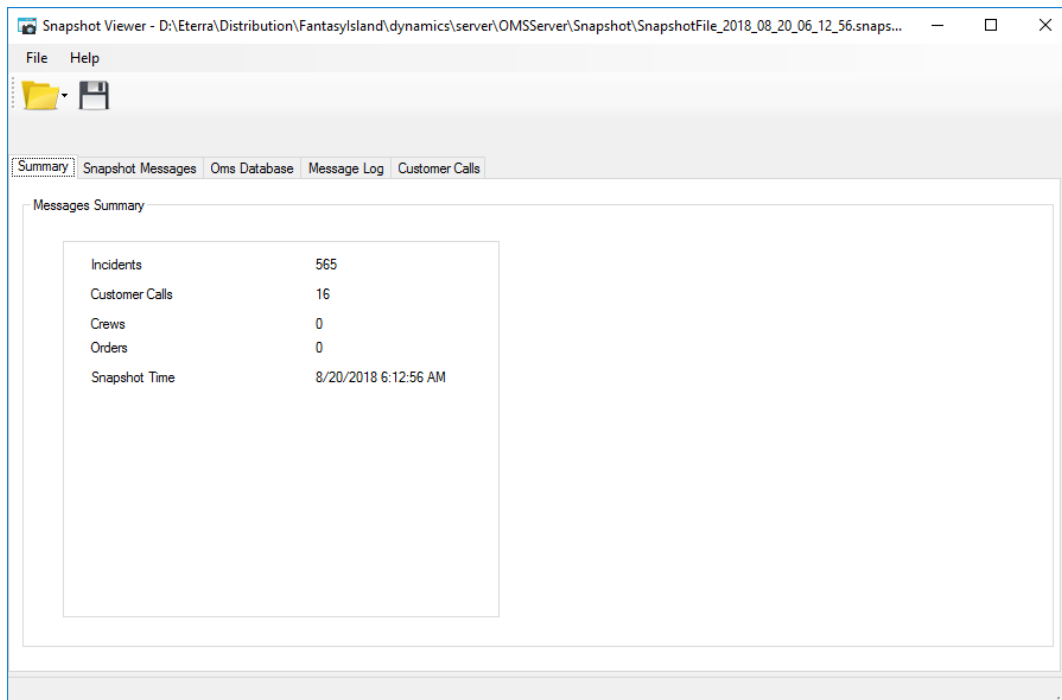


Figure 4. Viewing Common Snapshots - Summary Tab

2.4.1.2. Snapshot Messages Tab

The Snapshot Messages tab displays the messages in the opened snapshot.

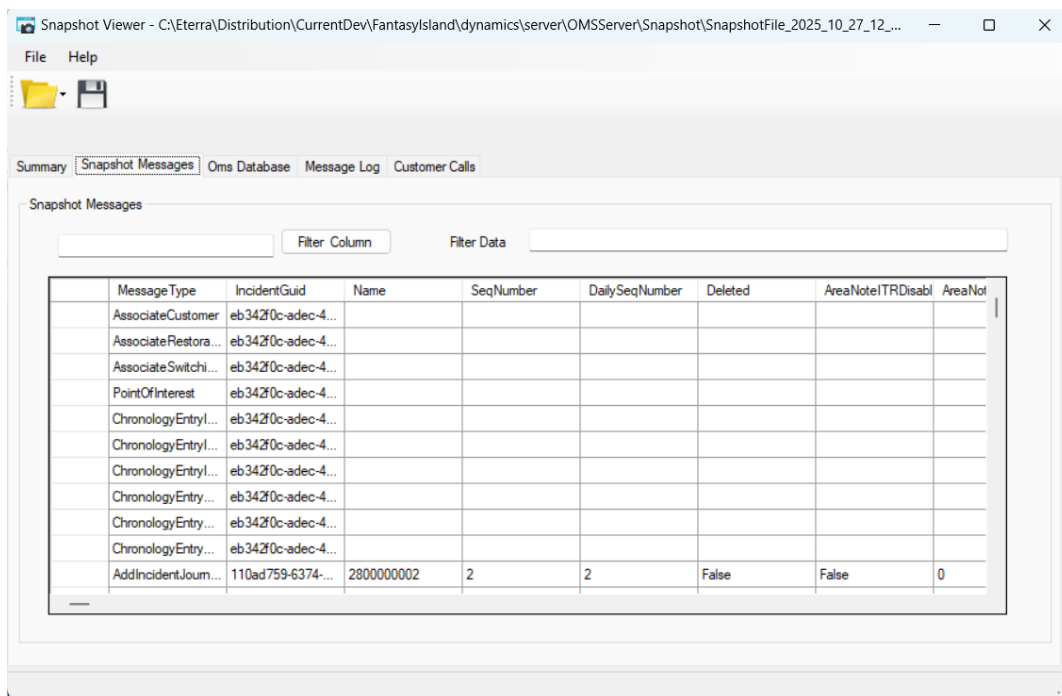


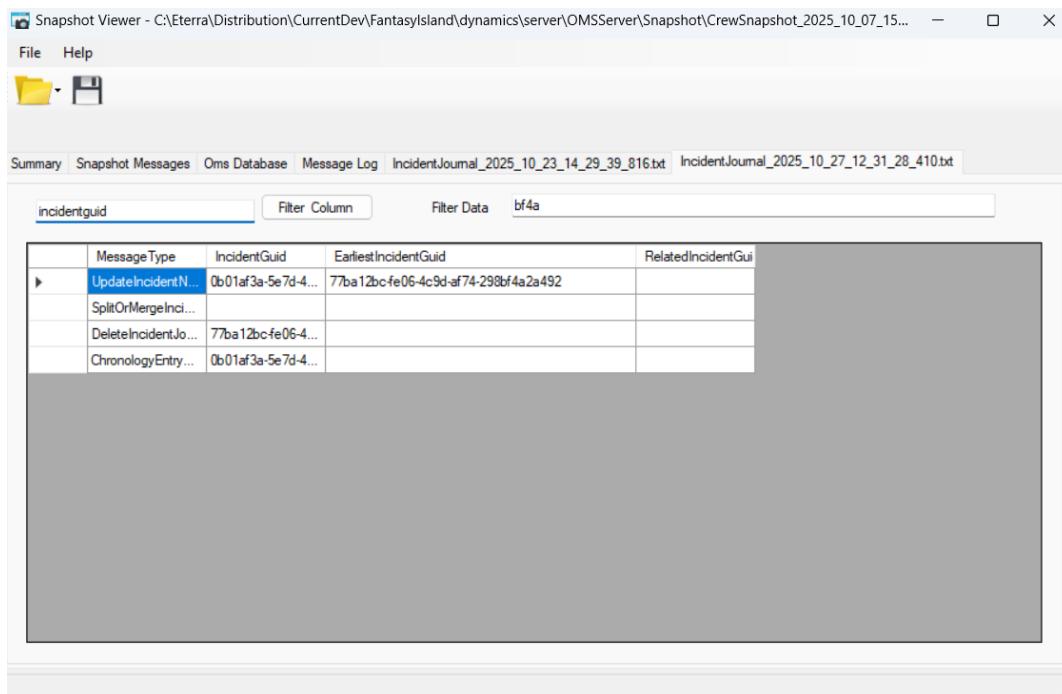
Figure 5. Snapshot Viewer, Snapshot Messages Tab

The following types of messages are displayed:

- AddIncidentJournalMessage
- AssociateServicePt
- AssociateCustomer
- AssociateRestorationStep
- AssociateSwitchingStep
- ChronologyEntryRestorationStep
- ChronologyEntryNtpSwitchOrderStep
- ChronologyEntryCreateIncident
- AreaNote
- GlobalIncidentJournalMessage

To filter data, some tabs provide two filter options:

- **Filter Column:** Filters the columns to display only those where the column name contains the typed word.
- **Filter Data:** Filters all data based on the typed word and displays all columns and fields that contain or exactly match it.



The Snapshot Messages table contains the same data as the Incident Journal file.

Note

The Snapshot Messages tab is populated only when viewing snapshots.

2.4.1.3. OMS Database Tab

The OMS Database tab shows the database state based on the opened snapshot and journal files.

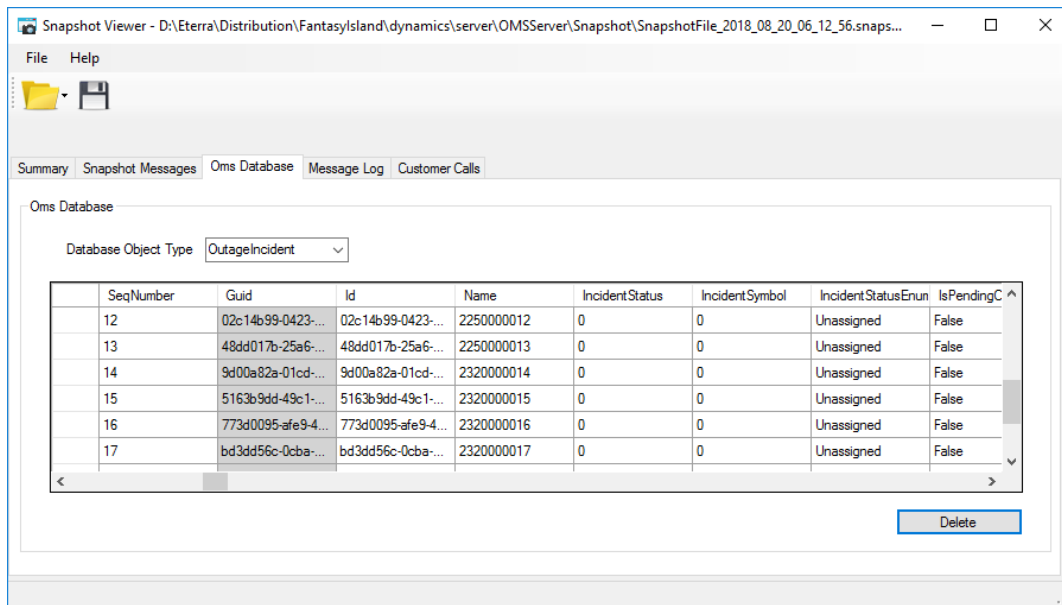


Figure 6. Snapshot Viewer, OMS Database Tab

You can select the following database object types:

- **OutageIncident:** Shows detailed information about registered incidents, such as sequence number, ID, location, outage type, outage and restoration time, associated equipment, and affected customers.
- **Crew:** Shows detailed information about registered crews, such as ID, employees, vehicles, status, operation area, location, and contact information.
- **Customer Status:** Shows detailed information about affected customers, such as customer key, energization status, AMI, and call info.

To delete the selected entry from the database, click the Delete button.

Note

The OMS Database tab is populated only when viewing snapshots.

2.4.1.4. Message Log Tab

The Message Log tab records error messages, user actions, and application activity.

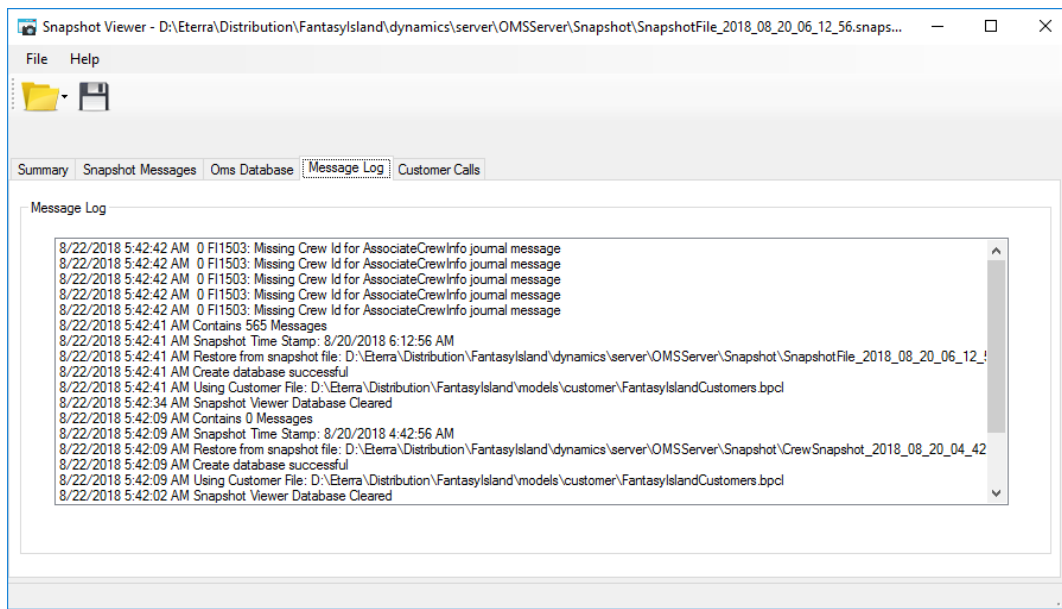


Figure 7. Snapshot Viewer, Message Log Tab

2.4.1.5. Customer Calls Tab

The Customer Calls tab provides details about recorded customers and unassociated calls.

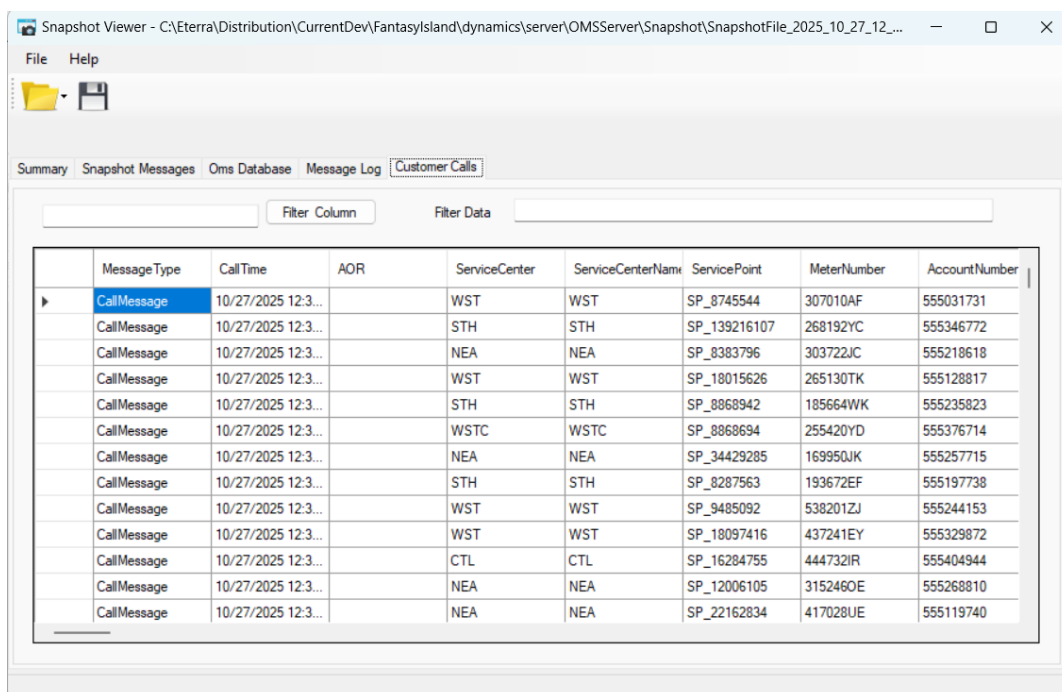


Figure 8. Snapshot Viewer, Message Log Tab

2.4.2. Viewing Journals

To view a journal file, click File > Open > Journal.

You can open Archive Journals, Call Journals, Call Server Journals, Crew Journals, Incident Journals, Real-Time Incident Journals, Order Journals, and Relay Server Journals. The default location of the folders containing journal files is %IDMS_ROOT%\dynamics\server\OMSServer.

Each journal file opens in a separate tab, as shown in the following image.

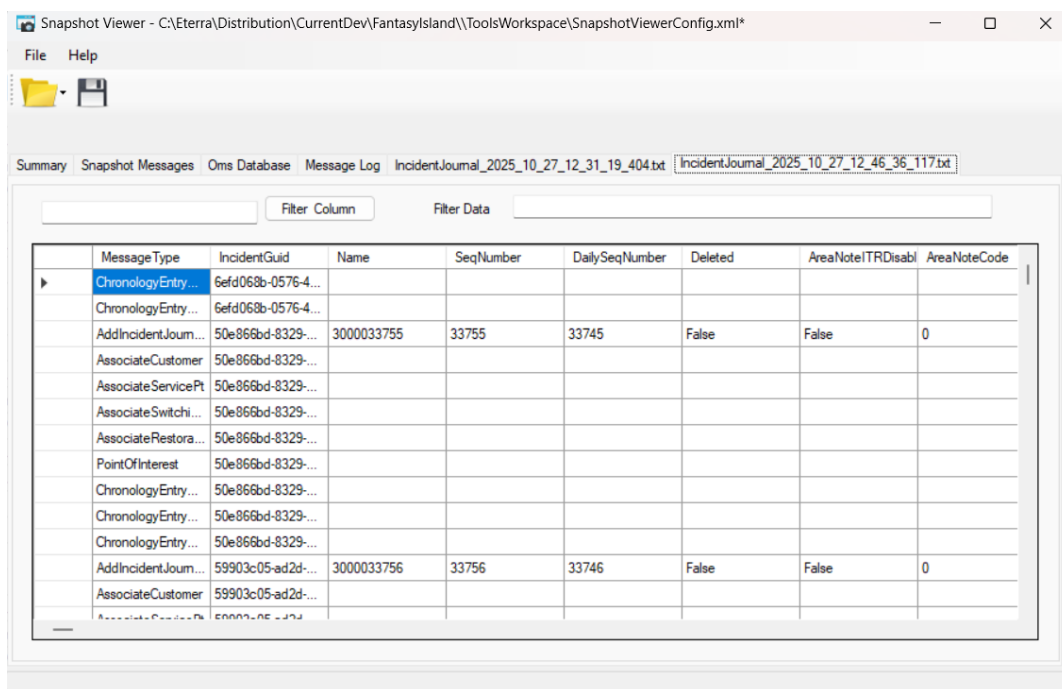


Figure 9. Viewing Journals

The journal messages that are suspected to contain errors are highlighted in yellow, and the suspected fields are highlighted in orange. The messages with modified data are highlighted in cyan.

By default, the client and server journal files are stored in the relevant folders under %IDMS_ROOT%\dynamics\server\OMSServer.

2.4.3. Viewing Dynamics

To view a dynamics file, click File > Open > Dynamics. By default, the client and server dynamics are stored in %IDMS_ROOT%\dynamics.

When a dynamics file is open, the dynamics summary appears on the Summary tab, and each type of dynamics appears on its own tab.

If you select the Show Only Affecting Entries check box, the Snapshot Viewer does not display overwritten entries. If you clear this check box, the Snapshot Viewer displays every entry from the

dynamic files, including the overwritten entries.

Note

By default, the read-only server overwrites entries in the dynamics.sav file. You can specify that the server not overwrite entries by setting the PRESERVE_ALL_DYNAMICS parameter to True (=1) in the server configuration file.

The Summary tab shows summary information related to the opened dynamics file.

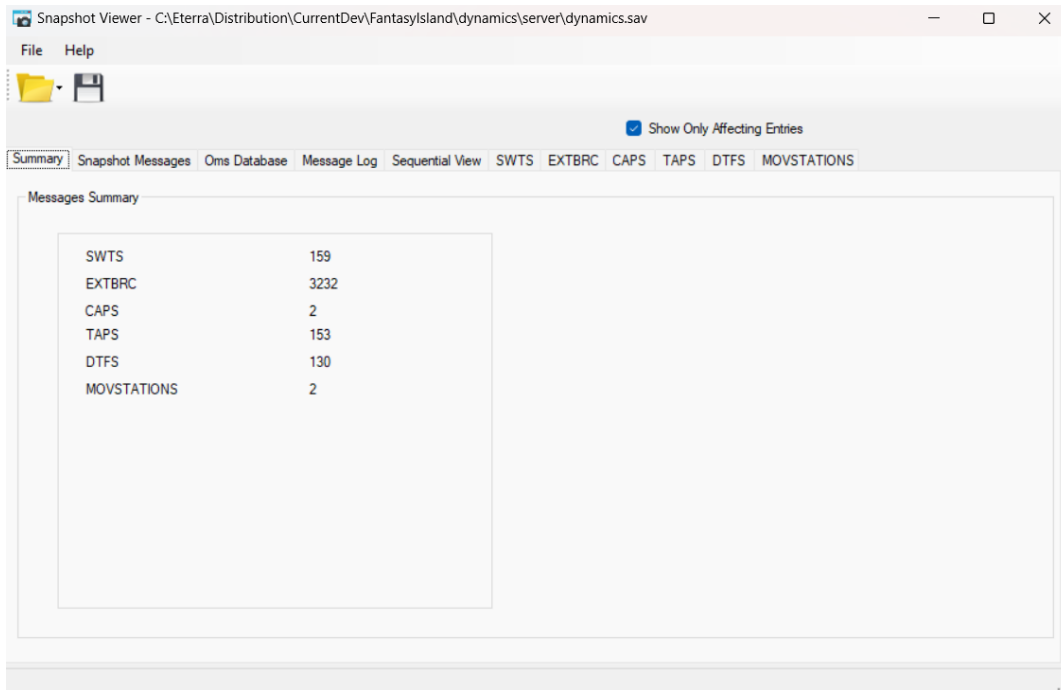


Figure 10. Viewing Dynamics - Summary Tab

The Messages Summary grid contains the following information:

- **SWTS:** Dynamics related to switches.
- **CUTS:** Dynamics related to cuts, if any.
- **EXTBRC:** Dynamics related to external branches.
- **JMPS:** Dynamics related to jumpers.
- **GRNDS:** Dynamics related to grounds.
- **TAGS:** Dynamics related to tags.
- **CAPS:** Dynamics related to capacitors.
- **COMPS:** Dynamics related to composite switches.
- **FIS:** Dynamics related to fault indicators.
- **TAPS:** Dynamics related to tap changers.
- **DTFS:** Dynamics related to distribution transformers.

- **MOVSTATIONS:** Dynamics related to mobile substations.
- **DECORATIONS:** Dynamics related to device state decorations.

To view dynamics related to particular devices, select the relevant tab.

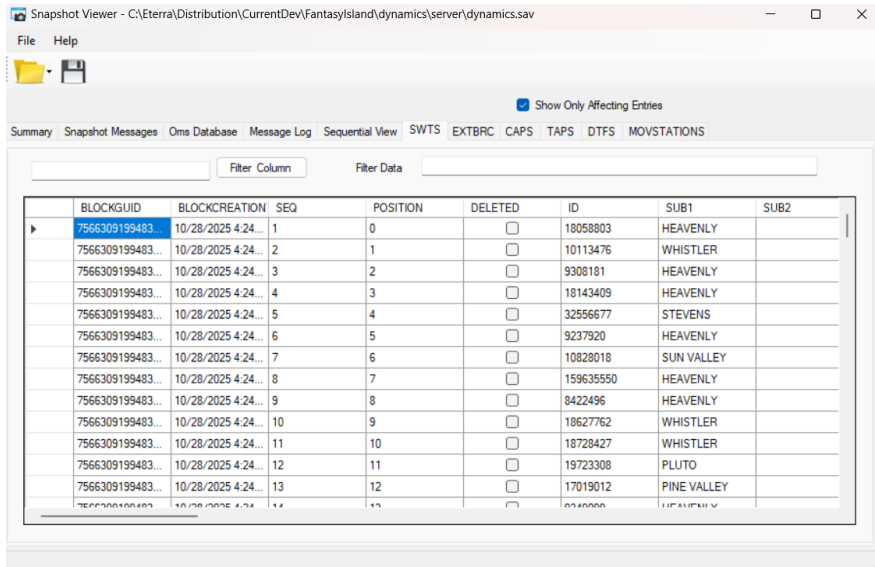


Figure 11. Viewing Dynamics - Device Tab

All dynamics have their own sequential number and position, which are summarized on the Sequential View tab.

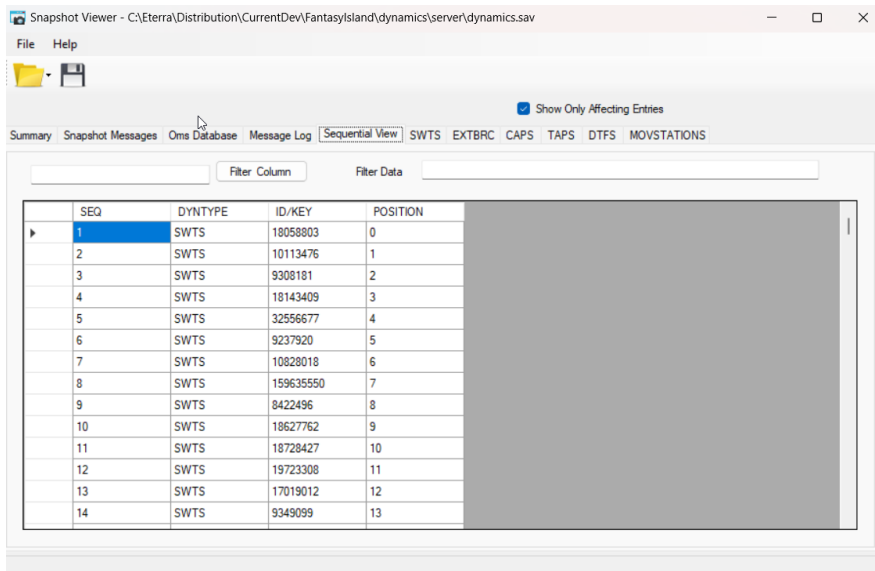


Figure 12. Viewing Dynamics - Sequential View Tab

3. Managing Dynamics Files

The Snapshot Viewer's capabilities include importing and exporting dynamics (.sav) files. The Snapshot Viewer supports the following file formats:

- TSV files, which contain data in a tab delimited format. You can import a dynamics (TSV) file into the Snapshot Viewer, review the data, and save the file in the (SAV) file format.
- SAV files, which are GE's proprietary binary file structure. You can open SAV files in the Snapshot Viewer and view the dynamics state of ADMS system components.
 - dynamics.sav files contain the current state of the system and are used by ADMS at the current time.
 - dynamics_YYYY_MM_DD.sav files contain the state of the system at midnight of MM/DD/YYYY.

To optimize file size, the dynamics files are compressed during the midnight backup. This compression includes removing all duplicated actions. Thus, only the last user action for each device is retained in the dynamics files.

The midnight compression can be done at run-time using the following command:

```
NETVIEW://RT/COMPRESSDYN
```

Note

You can disable the midnight compression by setting the `DISABLE_MID_NIGHT_DYNAMIC_COMPRESSION` parameter to True (=1) in the server configuration file.

- dynamics_YYYY_MM_DD_HH_MM_SS_bckp.sav files are backup files written periodically. The period (every 30 minutes, by default) is configured in the `DYNAMICS_BACKUP_MINUTES` parameter in the server configuration file.

You can view, edit, and save any of the files above in the Snapshot Viewer. You can alter the current state of the system by replacing the `dynamics.sav` file with a modified file.

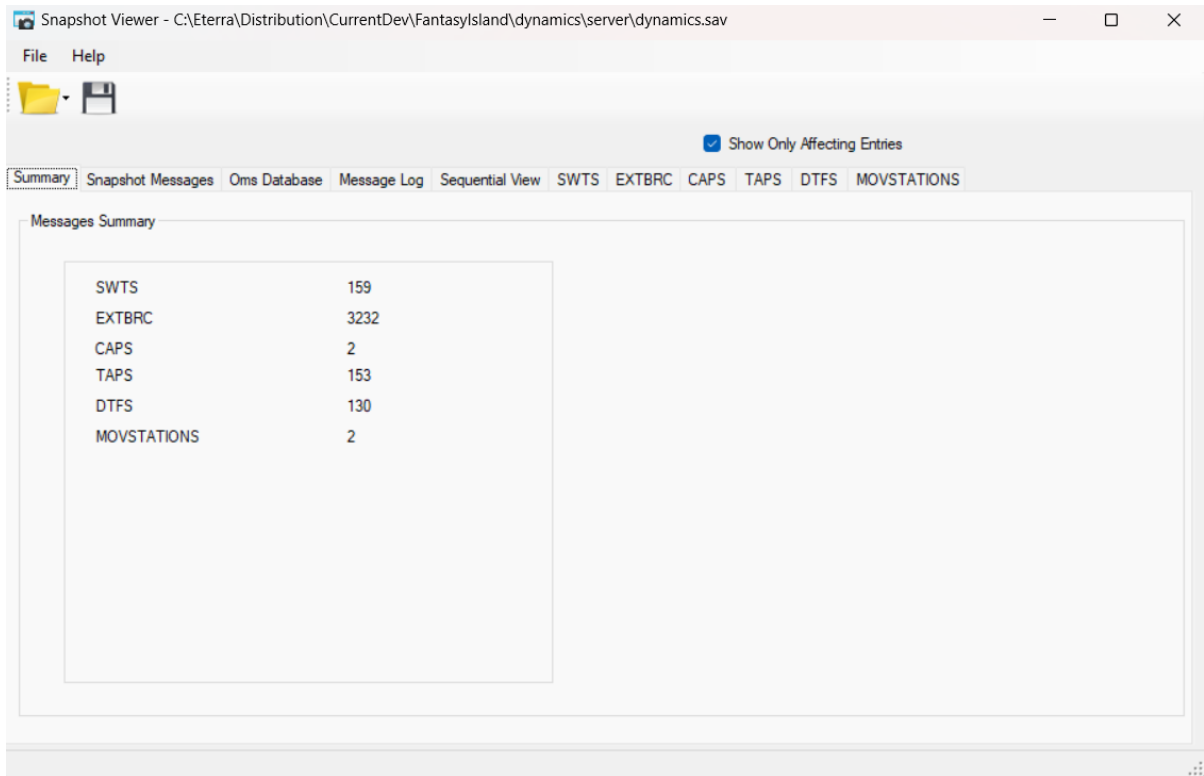
3.1. Exporting Dynamics Data from the ADMS Server

To export dynamics data from a SAV file into a TSV file for use by other applications:

1. Click File > Open Dynamics.
2. Navigate to the SAV dynamics data file.

The default location of the dynamics files is `%IDMS_ROOT%\dynamics\`.

Each dynamics type opens in its own tab. The Message Summary panel of the Summary tab is populated with data.



3. Click File > Export Dynamics > Data
4. In the Save As dialog box, enter a name for the TSV dynamics file, and click Save.

The dynamics data is exported in a tab-delimited format (TSV file). The export file contains both a header section and a data section:

- The header provides information about device/component types and field details for the data section.
- The data section contains the data for all abnormal devices and components.

The tool exports all data from the dynamics file (.sav) to a tab delimited format file that can be opened in a tool such as Microsoft Excel to facilitate data filtering and review.

3.2. Exporting Dynamics Headers

To export a dynamics header template from a SAV file for use in the import dynamics process:

1. Click File > Open Dynamics.
2. Navigate to the SAV dynamics data file.

The default location of the dynamics files is %IDMS_ROOT%\dynamics\.

Each dynamics type opens in its own tab. The Message Summary panel of the Summary tab is populated with data.



Note

Snapshot Viewer exports headers only for dynamic types that have entries. Therefore the SAV dynamics data file must contain at least one entry for each dynamics type to export a complete header template.

3. Click File > Export Dynamics > Header Template
4. In the Save As dialog box, enter a name for the SAV header template, and click Save.

The exported .sav file has all information about the fields and field data types for each dynamics type.

3.3. Importing Dynamics Files into the ADMS Server

To import dynamics into ADMS from a TSV file:

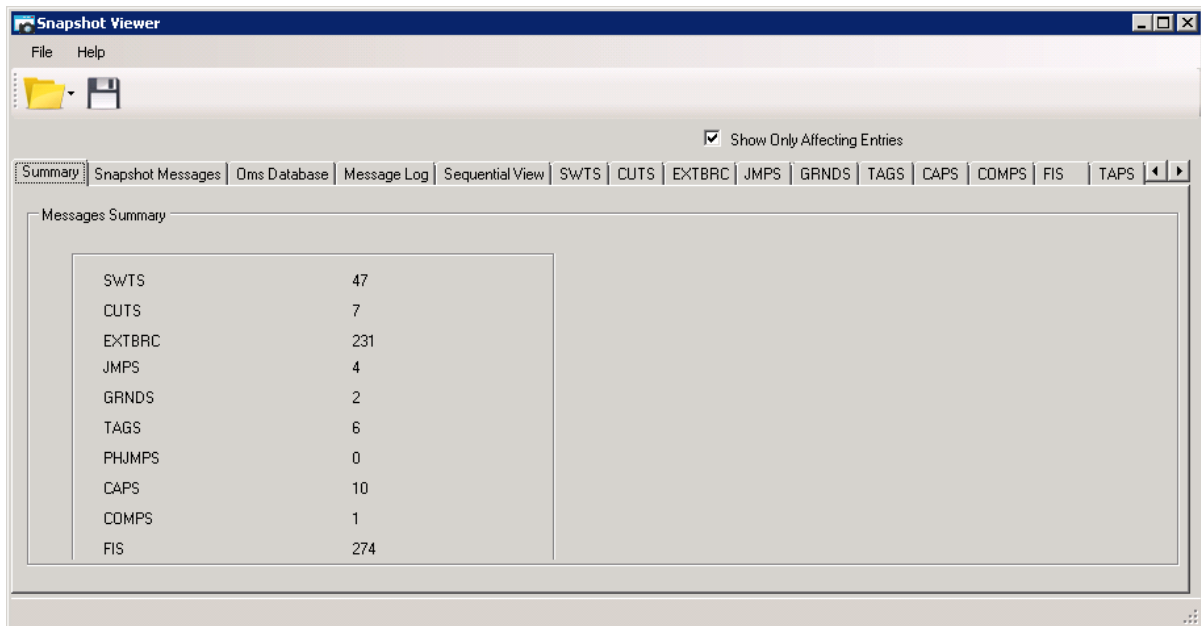
1. Click File > Import Dynamics > Header Template to upload the header template file.

You may upload an exported template (refer to [Exporting Dynamics Headers](#), or the default header template (Templatedynamics.sav), that is located in %IDMS_ROOT%\dynamics\server\OMSServer.

Uploading the header template file is required to recreate the custom binary SAV file. The TSV file does not contain such details as device or component type, field data type, and field size.

2. Click File > Import Dynamics > Data and navigate to the TSV dynamics data file.

Each dynamics type opens in its own tab and is populated with the dynamics data.



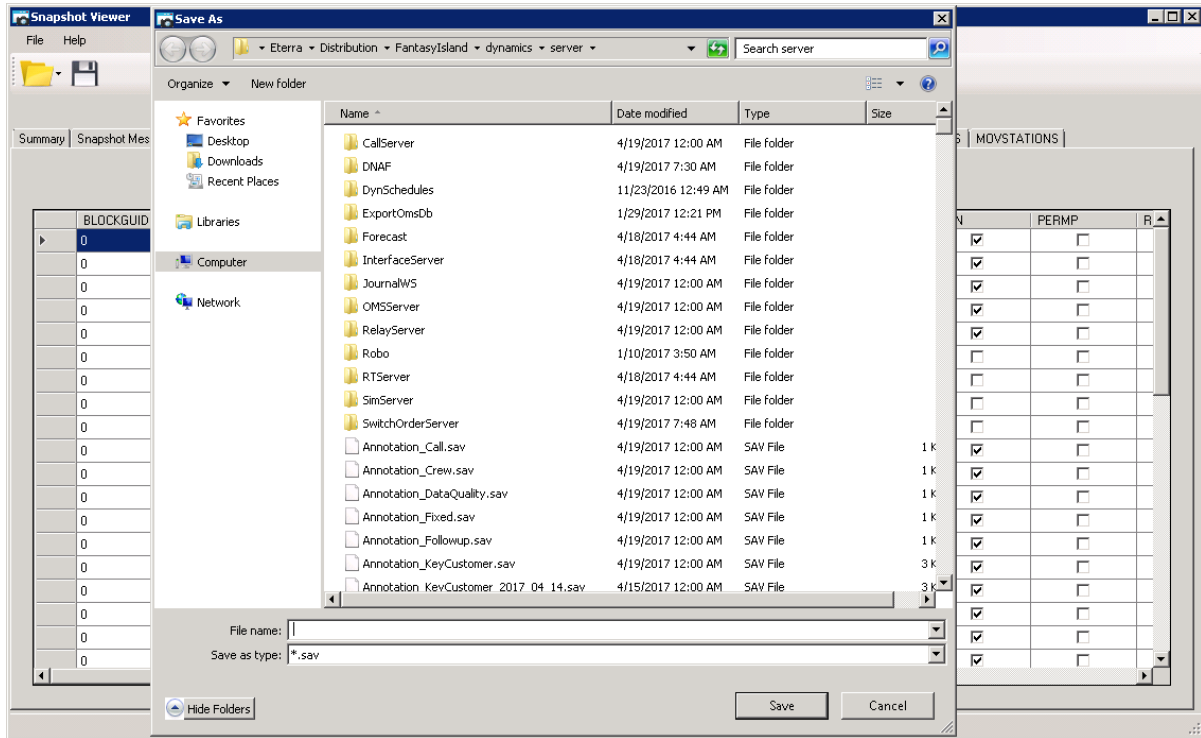
3. Review the data.

To filter the display to show only the affecting (latest) entries, select the Show Only Affecting

Entries check box. If you clear this check box, the Snapshot Viewer displays every entry from the dynamics file, including the overwritten entries.

4. Click File > Create Dynamics.

The Save As dialog box appears.



5. Enter a name for the new dynamics SAV file and then click Save.

The dynamics file is now saved in the SAV file format.

6. When the ADMS system is in production, the administrator may need to update the dynamics from an external system. To do so, repeat the all steps above for each TSV file to be imported and replace the ADMS Server dynamics files with these new SAV files.

Caution

A system dynamic file (*.sav, *.snapshot, *.journal) edited in the Snapshot Viewer tool should only be moved to the production system in a catastrophic situation and if no other options are available. In this case, the analyst must ensure that all the clients are logged off before the server is restarted with new dynamics.

3.4. Command-Line Interface

The Snapshot Viewer can manage dynamics files from the command prompt. The tool's command-line interface supports the following arguments.

/CONFIG [PATH TO FILE]

Specifies the path to a Snapshot Viewer configuration file. This argument is required when running the tool from the command line.

/IMPORT [PATH TO FILE]

Specifies the path to a TSV dynamics data file to be converted into a SAV dynamics data file.

/TEMPLATE [PATH TO FILE]

Specifies the path to a dynamics header template file (for example, `Templatedynamics.sav`) to be used for converting a SAV dynamics data file.

/OUTPUT [PATH TO FILE]

Specifies the path to a new SAV dynamics data file to convert from a TSV file.

/INPUT [PATH TO FILE]

Specifies the path to the input SAV dynamics data file to be exported into a TSV dynamics data file.

/EXPORT [PATH TO FILE]

Specifies the path to a new TSV dynamics data file to export dynamics data from a SAV file.

/SILENT

If this parameter is used, the Snapshot Viewer is working in silent mode (without user interface).

Exporting Dynamics Data from a SAV File Using the Command Prompt

You can export dynamics data from a SAV file automatically from the command prompt using the following command:

```
> SnapshotViewer.exe /CONFIG "configuration file path" /INPUT "SAV input file path" /EXPORT "export file path" -SILENT
```

For example, to export the `20170419_dynamics.tsv` file from the `dynamics.sav` file, in the command prompt enter the following:

```
D:\Eterra\Distribution\Tools\bin\SnapshotViewer.exe /CONFIG  
%IDMS_ROOT%\ToolsWorkspace\SnapshotViewerConfig.xml /INPUT %IDMS_ROOT%\dynamics\server\dynamics.sav  
/EXPORT %IDMS_ROOT%\dynamics\server\20170419_dynamics.tsv -SILENT
```

Converting SAV Files Using the Command Prompt

You can convert a SAV dynamics file automatically from the command prompt using the following command:

```
> SnapshotViewer.exe /CONFIG "configuration file path" /IMPORT "TSV input file path" /TEMPLATE "TSV header template file path" /OUTPUT "SAV output file path" -SILENT
```

For example, to convert the `dynamics.sav` file from the `20170419_dynamics.tsv` file, in the command prompt enter the following:

```
D:\Eterra\Distribution\Tools\bin\SnapshotViewer.exe /CONFIG
%IDMS_ROOT%\ToolsWorkspace\SnapshotViewerConfig.xml /IMPORT
%IDMS_ROOT%\dynamics\server\20170419_dynamics.tsv /TEMPLATE
%IDMS_ROOT%\dynamics\server\Templatedynamics.sav /OUTPUT %IDMS_ROOT%\dynamics\server\dynamics.sav
-SILENT
```

4. Managing Snapshots and Journals

The system includes the following types of snapshot files:

- SnapshotFile_YYYY_MM_DD_HH_MM_SS.snapshot,
CrewSnapshot_YYYY_MM_DD_HH_MM_SS.snapshot

NetServer writes these files at regular intervals as defined in the NetServer configuration file. The files contain data as of the writing time. The file names include a timestamp (for example, "SnapshotFile_2017_04_19_11_25_51").

Note

To create immediate snapshots of the current network state, open the NetServer window and select Create Snapshot from the Actions menu.

- DailySnapshot_YYYY_MM_DD.snapshot

NetDataServer creates this file once a day at midnight. The file contains all the data for the previous day.

In the Snapshot Viewer application you can view snapshot and journal file entries in a grid view, modify them, and save the changes. For NetServer and NetDataServer, you can apply journal messages up until a specified point in time and then create a new snapshot file with the saved results.

Both NetServer and the Snapshot Viewer open journal files for exclusive writing. The administrator must ensure that both applications do not run the same journal files simultaneously. The NetServer must be restarted once the files are corrected via the Snapshot Viewer.

The Snapshot Viewer application allows the administrator to open only one snapshot file at a time. After a snapshot file is opened, any number of journal files can be opened. This allows recreating the state of the OMS database at a certain point of time.

All journal files opened after the snapshot file query for messages after the time the snapshot was made. If only journals are opened and no snapshot file is open, queries into the journals display all messages from time zero.

The Snapshot Viewer does not automate any operations; the administrator must perform all the actions manually. The Snapshot Viewer workflow for reviewing the snapshots is as follows:

1. Open and review/modify the latest snapshot.
2. Open and review/modify the call and incident journal files. Look for yellow and orange highlights, which indicate errors.
3. Fix any errors in the journals.
4. Verify that the database solves the problem and is in the desired state.
5. Create a new snapshot that incorporates the changes.

4.1. Opening and Modifying Snapshot Files

To view snapshot files:

1. From the File menu, select Open > Snapshot.

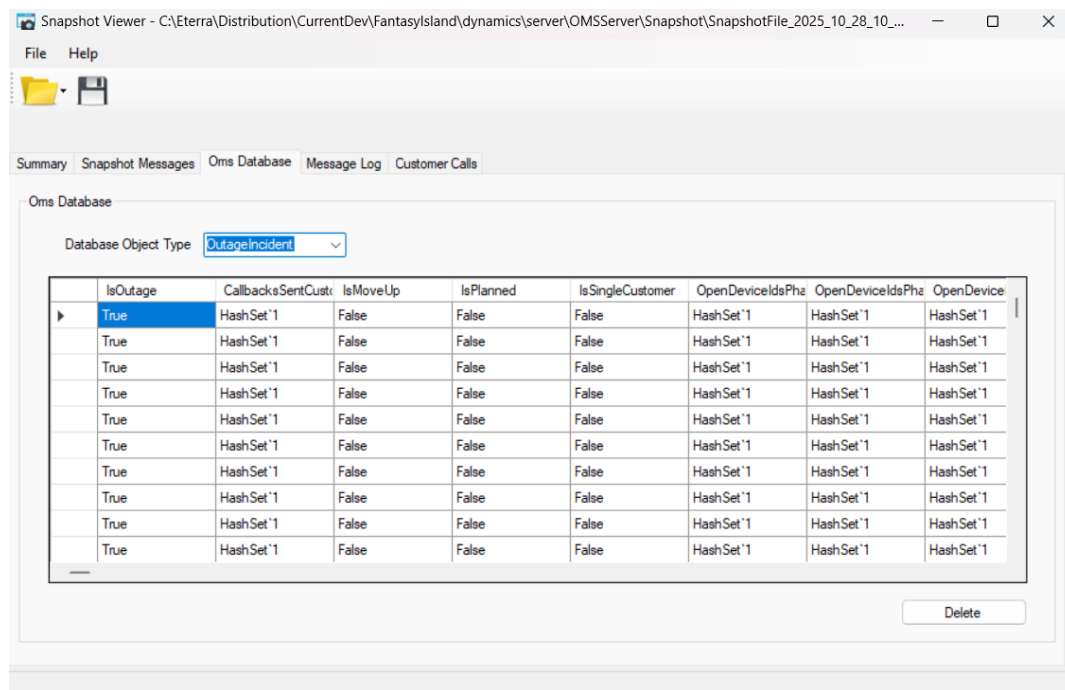
The Open dialog box appears.

2. Browse to the snapshot file location, and click Open.

The default location is %IDMS_ROOT%\dynamics\server\OMSServer\Snapshot.

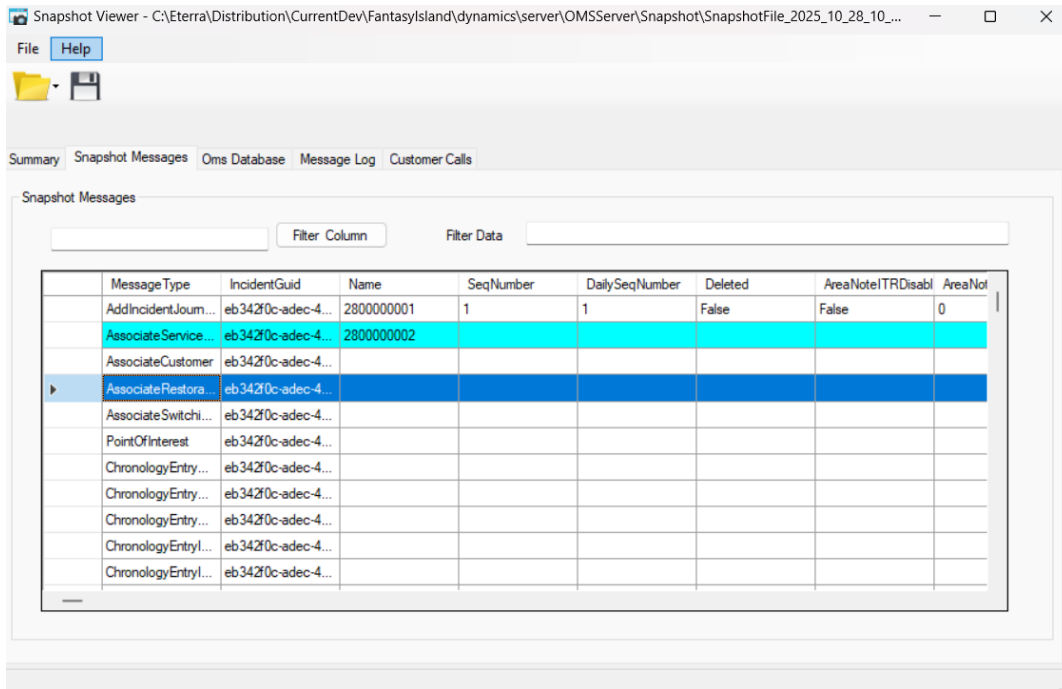
The snapshot file opens. The Message Log tab displays the file's timestamp and the number of messages in the file.

3. Review the data on the Summary tab and the Snapshot Messages tab.
4. On the OMS Database tab, filter the displayed data by selecting a Database Object Type.



5. To edit data, on the Snapshot Messages tab, click a field and modify the information.

The modified messages are highlighted in cyan.



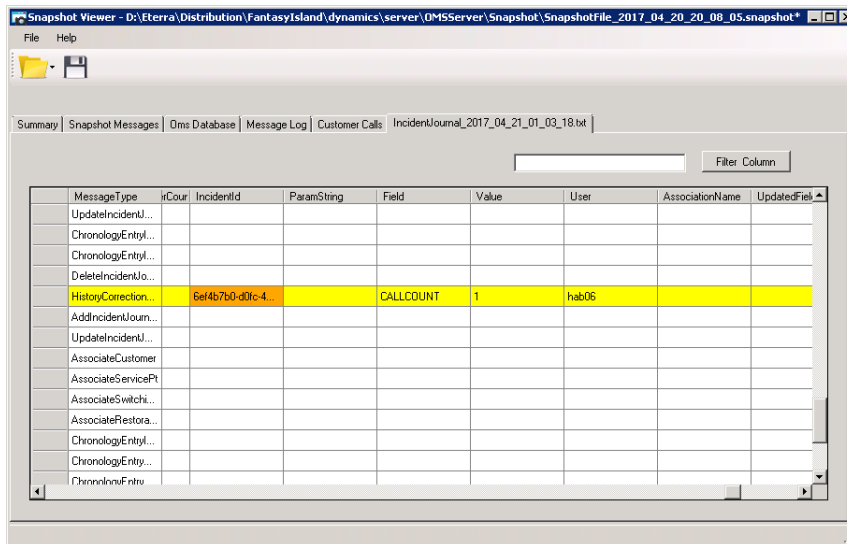
4.2. Opening and Modifying Journals

To view and modify call and incident journals

1. From the File menu, select Open >Journals.
2. Open incident journals and call journals in sequence starting from the snapshot time. The default location of the folders containing journal files is %IDMS_ROOT%\dynamics\server\OMSServer.

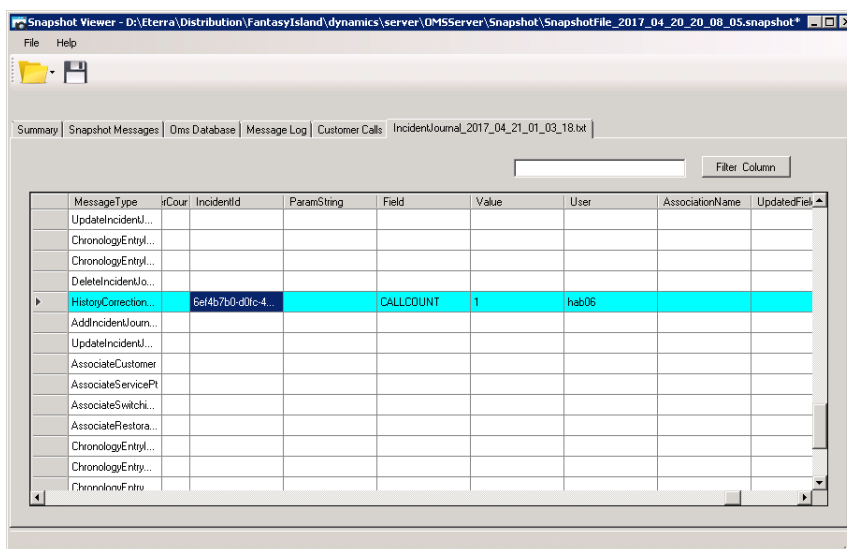
Each journal opens in a separate tab.

Suspected errors in the journal messages are highlighted in yellow. Columns are highlighted in orange.



3. Edit the messages to fix any errors and solve the original problem.

Cyan highlight indicates that the message was updated.



4.3. Creating a New Snapshot

To create a new snapshot:

1. Open a snapshot.
2. Review the edited database to verify that the problem is solved and the database is in the desired state.
3. From the File menu, click Create Snapshot.

The Save As pop-up window appears.

4. Enter the snapshot file name and click Save.

The file is saved with the ".snapshot" extension.

i Note

Even if the new snapshot file is created under the name of the original snapshot file, the modifications in this file are time stamped. Therefore, the database is not saved or saved under a new name, but rather a new instance of the database is created.